BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 164 · Final GCSE Computer Science Consolidation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What should guide the choice of a final revision topic?

Q2. How many colours can a bitmap with 4 bits per pixel use?
A. 4
B. 8
C. 32
D. 16

Q3. What happens to raw size when both bitmap dimensions double at fixed depth?

Q4. A microphone records digital sound. “Sampling stores every instant of the sound wave.” Correct this. How does it affect recorded detail?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

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I Do · Read new knowledge

Key vocabulary

Recall the full course — Paper 1 systems and Paper 2 computational thinking are equal components.

Use accurate language — State precise definitions and causal effects.

Demonstrate independence — Complete unseen traces, calculations and explanation without hints.

Check answers — Use command words, units, bounds and allocated marks before submission.

New learning

Recall the full course: Paper 1 systems and Paper 2 computational thinking are equal components.

Use accurate language: State precise definitions and causal effects.

Demonstrate independence: Complete unseen traces, calculations and explanation without hints.

Check answers: Use command words, units, bounds and allocated marks before submission.

Worked example

Solve an integrated question: a 20×20 bitmap with 8-bit depth is uploaded over a network. Calculate uncompressed bytes and explain one network factor.

1. Pixels =20×20=400.

2. Bits =400×8=3200.

3. Bytes=3200÷8=400.

4. Bandwidth, latency or congestion can affect upload time; explain the chosen factor.

We Do · Practise with guidance

Guided 1

A list [2,6,10,14,18] is sorted. Trace binary search for 18.

Worked method: Found at index4.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Print the combined size of the first and last file in this list.

```
array filesMB = [2,6,10]
print(...)
```

write

Question 1

Give one difference between compiler and interpreter. [3 marks]

Question 2

Calculate 10101010 binary in denary. [2 marks]

Question 3

Explain how to make a program robust against invalid numeric input. [4 marks]

Exit Ticket · Answer before checking

OCR J277 has two equally weighted examination components.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Evidence of a specific knowledge or method gap from the recent targeted review.

Q2. D — 16

Q3. The number of pixels and raw pixel-data size become four times as large.

Q4. Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured.

I Do / We Do pages · Guided practice

Q1. Found at index4.

You Do pages · Code First

```
array filesMB = [2,6,10]
print(filesMB[0] + filesMB[2])
```

Expected result: 12

You Do pages · Independent questions

Q1. Compiled source is translated to executable/intermediate output before execution of translated program; interpreted execution is mediated by an interpreter.

Q2. 170.

Q3. Use type/range checks; handle failure safely and request another value; test boundary/erroneous cases.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: